

Problem 1 — Pareto Efficiency

1. Minimum criterion for a just society

From a welfare-economics perspective, the minimal criterion is Pareto efficiency: an allocation should be such that no one could be made better off without making someone else worse off. It's a deliberately weak, minimal bar — it says nothing about fairness or how welfare should be distributed, only that no resources or potential gains are being needlessly wasted.

2. What Pareto efficiency measures

It measures allocative efficiency — whether resources are allocated so that no further reallocation could make someone better off without harming someone else. It does not measure equity, fairness, or how welfare is distributed across people.

3. When is an allocation Pareto-inefficient?

When a feasible alternative allocation exists that would make at least one person better off without making anyone else worse off — i.e., a Pareto improvement is still available and hasn't been taken.

4. Why Pareto efficiency is an intuitive justice criterion

Because it commands unanimous agreement: everyone can agree that leaving a mutually beneficial improvement on the table (helping someone without hurting anyone) is undesirable. It requires no controversial interpersonal comparisons of utility or value judgments about whose welfare should count more — making it the most minimal, widely acceptable notion of social improvement available.

5. Which grading scenario is "just"?

Since the assumption is that a classmate's grade has no bearing on your own utility, Scenario 2 (you: 5.0, friend: 5.5) is a Pareto improvement over Scenario 1 (5.0, 5.0) — you are exactly as well off, and your friend is strictly better off. By the Pareto criterion alone, Scenario 2 is therefore "better" (Pareto-superior). This is a useful illustration of the limits of Pareto efficiency as a justice criterion (see part 8): most people's intuitive sense of fairness objects to the lecturer favoring your friend for arbitrary reasons, even though Pareto efficiency has nothing to say against it, since no one's welfare was reduced.

6. What is a Pareto improvement?

A change in allocation that makes at least one person strictly better off while making no one else worse off.

7. Table 5.1 — which scenarios are Pareto-efficient / Pareto improvements?

	Your grade	Friend's grade
Scenario 1	5.0	5.0
Scenario 2	5.0	5.5
Scenario 3	6.0	4.5

Scenario 1 → Scenario 2 is a Pareto improvement (you're unchanged, your friend gains).

Scenario 1 → Scenario 3 is not a Pareto improvement (you gain, but your friend loses). Since Scenario 2 Pareto-dominates Scenario 1, Scenario 1 is not Pareto efficient. Scenario 2 and

Scenario 3 cannot be ranked against each other by the Pareto criterion — each makes one person better off and the other worse off relative to the other, so the Pareto criterion is silent between them. As neither is dominated by a feasible alternative, both Scenario 2 and Scenario 3 are Pareto efficient; only Scenario 1 → Scenario 2 qualifies as a Pareto improvement among the three.

8. Main critique of Pareto efficiency

It is silent on distribution and equity. Pareto efficiency can validate highly unequal allocations as "efficient" (an allocation where one person has everything and another has nothing can be Pareto efficient, since redistributing would make the first person worse off), and — as in part 7 — it cannot rank many pairs of allocations at all whenever a change benefits one person at another's expense. It is therefore a very weak notion of social welfare that says nothing about fairness.

Problem 2 — Normative Theories

1. Three classes of normative theories

Consequentialism, deontology, and virtue ethics. Welfarism — the view that social states should be judged solely by their consequences for individual well-being — belongs to consequentialism. Its main normative premise: what matters morally is only the aggregate pattern of well-being (utility) that results from an action or policy, not the nature of the actions, rules, rights, or character behind it — "the ends (outcomes) are what count."

2. Table 5.2 — normative premises and take on institutions

	Normative premise	Take on institutions
Consequentialism	The moral value of an action or state depends only on its consequences (e.g., total welfare produced).	Purely instrumental: institutions (markets, laws) have no intrinsic value — they're judged solely by whether they produce good outcomes.
Deontology	Certain acts, rules, or rights are inherently right or wrong regardless of consequences (duties must be respected).	Institutions matter because they embody and enforce rights/duties (e.g., property rights, contracts, due process), independent of whether doing so maximizes aggregate welfare.
Virtue ethics	Moral value flows from the character/virtues of the agent (honesty, fairness, courage), not from calculating outcomes or following fixed rules.	Institutions are judged by whether they cultivate virtuous character and dispositions (trust, cooperation) in participants, not by outcomes or rule-compliance alone.

3. Normative theory used in mainstream economics

Consequentialism, and specifically welfarism — mainstream (welfare) economics evaluates policies and institutions by their consequences for aggregate welfare (total surplus, Pareto efficiency), largely setting aside considerations of rights, duties, or character.

4. Mainstream economists' take on institutions vs. virtue ethics

Mainstream economists treat institutions purely instrumentally: a market or policy is good only insofar as it produces efficient/welfare-maximizing outcomes, with no value placed on the institution itself. This differs sharply from virtue ethics, which would judge an institution by whether it cultivates virtuous character and dispositions in the people who participate in it — a criterion entirely separate from measured outcomes or efficiency.

Problem 3 — Consumer & Producer Surplus

1. Interpreting points along the demand function (Fig. 5.1)

Each point on the demand curve shows the maximum price a consumer is willing to pay for one additional (marginal) unit at that quantity — i.e., the marginal willingness to pay. Reading the curve from left to right, it also shows how many units would be bought at each possible price; points further left/up represent units that buyers value more highly than the units further right/down.

2. Interpretation of consumer surplus

Consumer surplus is the difference between what consumers are willing to pay for each unit (their marginal willingness to pay, from the demand curve) and what they actually pay (the market price), summed across all units purchased. It measures the net benefit consumers get from being able to buy at the market price rather than at their own (higher) maximum willingness to pay.

3. Calculate consumer surplus: $P(x) = 100 - \frac{1}{2}x$, market price $p = 50$

At $p = 50$: $50 = 100 - 0.5x \Rightarrow x = 100$. Consumer surplus is the triangle between the demand curve and the price line, from 0 to $x = 100$:

$$CS = \frac{1}{2} \cdot \text{base} \cdot \text{height} = \frac{1}{2} \cdot 100 \cdot (100 - 50) = \frac{1}{2} \cdot 100 \cdot 50 = 2,500$$

4. Interpreting points along the supply function (Fig. 5.2)

Each point on the supply curve shows the minimum price a producer must receive to be willing to supply one additional (marginal) unit — i.e., the marginal cost of that unit. Reading left to right, it shows the quantity producers are willing to supply at each price.

5. Interpretation of producer surplus

Producer surplus is the difference between the price producers actually receive (the market price) and the minimum price they would have accepted for each unit (their marginal cost, from the supply curve), summed across all units sold. It measures the net benefit producers get from selling at the market price rather than at their own (lower) minimum acceptable price.

6. Calculate producer surplus: demand $P(x)=100-\frac{1}{2}x$, supply $Q(y)=\frac{1}{2}y$, $p=50$

At $p = 50$: $50 = 0.5y \Rightarrow y = 100$ (note this matches quantity demanded, confirming $p = 50$, $q = 100$ is the market equilibrium). Producer surplus is the triangle between the price line and the supply curve, from 0 to $y = 100$:

$$PS = \frac{1}{2} \cdot \text{base} \cdot \text{height} = \frac{1}{2} \cdot 100 \cdot (50 - 0) = 2,500$$

7. Aggregate surplus

$$\text{Aggregate surplus} = CS + PS = 2,500 + 2,500 = 5,000$$

Aggregate (total) surplus indicates the total net gain to society — consumers and producers combined — from exchange taking place in this market. It is a measure of the total value created by the market, and it is maximized precisely at the competitive equilibrium price and quantity (this is the substance of the First Theorem of Welfare Economics, covered in Problem 4).

Problem 4 — Theorems of Welfare Economics

1. First Theorem — implication for consumer and producer surplus

The First Theorem states that every competitive equilibrium (under standard conditions: complete markets, no externalities, perfect information, price-taking behavior) is Pareto efficient. Its implication for surplus: the competitive equilibrium price and quantity maximize aggregate surplus ($CS + PS$) — any deviation from that price/quantity (e.g., a price ceiling, floor, or quota) reduces total surplus, creating a deadweight loss.

2. Argument for or against a market economy?

It is generally read as an argument for market economies: it shows that decentralized trading among self-interested, price-taking agents — with no central planner — arrives at a Pareto-efficient allocation (the "invisible hand" result). But it is a conditional argument: it only holds when the theorem's assumptions hold. Where they fail (externalities, market power, incomplete information), unregulated markets are not guaranteed to be efficient, so it is not an unconditional case for laissez-faire.

3. Second Theorem — implication for the distribution of welfare

The Second Theorem states that any Pareto-efficient allocation can be achieved as a competitive equilibrium, provided the initial endowments are redistributed appropriately (via lump-sum transfers) before trading occurs. Its implication: efficiency and equity can, in principle, be separated — society can select any distribution of welfare it considers fair (any point on the Pareto frontier) and still rely on competitive markets to reach it efficiently, simply by redistributing initial resources beforehand rather than intervening in prices or allocations directly.

4. How markets guide resources to their most efficient uses

Through the price mechanism: prices aggregate dispersed information about scarcity, cost, and consumer preferences. Profit-seeking firms and utility-maximizing consumers respond to these price signals by shifting resources toward their highest-valued uses — without any central coordinator — and this decentralized process converges toward the efficient allocation described by the First Theorem.

5. Caveats: incentives, enforcement, and decentralization

- Incentives: the Second Theorem requires lump-sum transfers — redistribution that doesn't depend on or distort people's choices. Real-world redistribution (income taxes, subsidies) is rarely lump-sum; it typically depends on income or consumption choices, creating distortionary incentive effects (e.g., taxing income discourages work) that undercut the clean efficiency/equity separation the theorem assumes.
- Ability to enforce redistribution: implementing the "correct" initial redistribution requires observing people's true endowments, abilities, or needs and enforcing transfers accordingly — information asymmetries make this difficult in practice, limiting how well the theorem's redistributive prescription can actually be carried out.
- Decentralization: the theorems' appeal rests on decentralized decision-making by individual agents responding to prices — yet achieving the Second Theorem's ideal redistribution requires some centralized authority to determine and enforce it, creating a tension between the efficiency benefits of decentralization and the centralized intervention needed to pursue equity goals.

Problem 5 — Revealed Preference

1. Main conjecture and its implications

The revealed-preference approach assumes that people's actual choices directly reveal their true underlying preferences — individuals are taken to be the best judges of their own welfare, so observed behavior is, by definition, what is best for them. Implication: welfare can be inferred directly from observed choices (e.g., market demand), without needing an outside standard of what's "really" good for someone.

2. Implications of accepting vs. rejecting the approach

If accepted: welfare analysis can rely straightforwardly on observed choices (as in the consumer-surplus calculations of Problem 3), policy should generally respect individual choices (consumer sovereignty), and paternalistic intervention is hard to justify. If rejected: choices can no longer be assumed to reveal true welfare — people may err, act on biases, or

work against their own interests — so choice-based welfare measures may overstate true welfare, and there may be a case for corrective or paternalistic policies (e.g., nudges, restrictions, mandatory savings) that improve well-being beyond what raw choices would suggest.

3. Examples where individuals may not act in their own best interest

- Addiction (smoking, gambling, drugs) — choices driven by compulsion rather than considered judgment.
- Self-control / present-bias problems — under-saving for retirement, procrastination, unhealthy eating despite intending otherwise.
- Decisions made under strong emotion, stress, or misinformation — impulse purchases, scams.
- Choices made by children or individuals with limited information or cognitive capacity.

4. Consequences for the welfare theorems if the approach is rejected

The welfare theorems rely on utility-maximizing choices being a faithful guide to true well-being. If choices no longer reliably reveal genuine preferences, a competitive equilibrium can be "efficient" in the narrow sense of matching people's observed (possibly mistaken) choices, while leaving people worse off in terms of actual well-being (e.g., an addicted consumer's purchases are formally utility-maximizing but don't reflect genuine welfare). This undermines the normative force of the First and Second Welfare Theorems as arguments that market outcomes are desirable.

Problem 6 — City Consultant: Local Emissions

1. Interdependencies, externalities, and internalization

An interdependency exists whenever one party's actions affect another party's well-being or production (e.g., a factory's emissions affecting nearby residents). When such an interdependency isn't reflected in any market price — no one pays or is compensated for it — it's called an externality: the affected party bears a cost (or gets a benefit) the decision-maker ignores. Internalization is the process of making the responsible party account for that externality in their own decisions — through a tax equal to the external cost, an assignment of property rights and bargaining, or regulation — so private incentives line up with true social costs and benefits.

2. Are emissions causing efficiency problems for the city?

Yes. Firms bear only their own private production costs, not the health and quality-of-life costs imposed on nearby residents, so they will produce (and pollute) more than the socially efficient level — a textbook negative externality and market failure, since the marginal social cost of production (private cost plus external cost) exceeds the marginal private cost the firms actually respond to.

3. Can allocating property rights achieve efficiency?

If property rights (e.g., a right to clean air, or a right to emit) are clearly assigned and transaction costs are low, the Coase theorem suggests private bargaining between firms and residents can reach an efficient outcome regardless of who holds the rights (this is illustrated numerically in Problem 7). In practice this is limited here: local emissions affect many dispersed residents, so coordinating a bargain runs into a free-rider/coordination problem among the residents themselves, and damages are often hard to measure — so relying purely on property rights and private bargaining is unlikely to fully solve the problem without some facilitating role for the city.

4. Would a uniform per-firm emissions limit be efficient?

Generally not. Firms differ in how costly it is for them to reduce emissions (abatement cost). An efficient reduction requires low-abatement-cost firms to cut back more and high-abatement-cost firms to cut back less, equalizing marginal abatement cost across firms for any given total reduction. A uniform cap applied equally to every firm ignores these cost differences, achieving any given total reduction at a higher cost than necessary.

5. Cap-and-trade market for emission certificates — how does it differ from Question 3?

A cap-and-trade system centrally sets the total (aggregate) emissions cap — addressing the coordination problem that made pure Coasean bargaining difficult among many dispersed residents — and then lets firms trade permits among themselves. Firms with low abatement costs sell surplus permits; firms with high abatement costs buy extra permits; trading continues until marginal abatement costs are equalized across all firms. This combines a government-set efficient total quantity with market-based efficiency in determining who actually reduces emissions, at the lowest total cost.

6. What changes if emissions are global rather than local?

The set of affected parties expands from local residents to essentially the whole world, across generations — making Coasean bargaining practically impossible (transaction costs are prohibitive with billions of affected parties across many legal systems, and no global authority can assign or enforce the necessary property rights). A single city's or country's regulation also becomes insufficient on its own, since one jurisdiction's emissions cuts benefit everyone globally, creating a global free-rider problem where each country is tempted to let others bear the cost of reducing emissions. Effective solutions generally require international cooperation (treaties, global carbon pricing) rather than purely local property-rights or regulatory fixes.

Problem 7 — Silentium and Gaudium

Silentium guests	Silentium profit (no externality)	Gaudium guests	Gaudium profit
0	0	0	0
10	10,000	100	50,000
20	12,000	200	60,000
30	13,000	300	65,000
40	11,000	400	66,000

1. Is there an externality?

Yes — Gaudium's noise reduces Silentium's profit without Gaudium bearing any of that cost itself (absent a rights regime). This is a one-directional negative externality caused by Gaudium.

2. Silentium's actual profit as a function of guests on both islands

Actual Silentium profit = (baseline profit at its own guest count) – 20 × (Gaudium's guests):

Silentium \ Gaudium guests →	0	100	200	300	400
0	0	-2000	-4000	-6000	-8000

10	10000	8000	6000	4000	2000
20	12000	10000	8000	6000	4000
30	13000	11000	9000	7000	5000
40	11000	9000	7000	5000	3000

3. Combination maximizing total profits for both hotels

Total profit = [Silentium's baseline profit(g_S) – $20 \cdot g_G$] + Gaudium's profit(g_G). Since Silentium's own choice and Gaudium's choice enter separately, each can be optimized on its own: Silentium's baseline profit is maximized at $g_S = 30$ guests (13,000). Gaudium's profit net of the externality it causes, $\pi_G(g_G) - 20 \cdot g_G$, is maximized at $g_G = 300$ guests (65,000 – 6,000 = 59,000; compare: $g_G=100 \rightarrow 48,000$, $g_G=200 \rightarrow 56,000$, $g_G=400 \rightarrow 58,000$).

Efficient outcome: Silentium hosts 30 guests, Gaudium hosts 300 guests.

Silentium's profit = 13,000 – 6,000 = 7,000. Gaudium's profit = 65,000. Total = 72,000.

4. Silentium holds the right to demand compensation for losses

Gaudium must now fully compensate Silentium (20 per guest) for the externality it causes, so Gaudium internalizes the cost and chooses g_G to maximize $\pi_G(g_G) - 20 \cdot g_G$ — the same calculation as part 3 — landing again at $g_G = 300$. Silentium, now fully insulated by compensation, still chooses $g_S = 30$ to maximize its own baseline profit.

Guests: Silentium = 30, Gaudium = 300 — identical quantities to Question 3.

Silentium's profit = 13,000 (fully compensated). Gaudium's profit = 65,000 – 6,000 = 59,000. Total = 72,000.

Only the distribution of profit changes relative to Question 3 (Silentium now nets the full 13,000 instead of 7,000, Gaudium nets 59,000 instead of 65,000) — the efficient quantities themselves are unchanged. This is exactly the Coase theorem at work: assigning the property right to the harmed party doesn't change the efficient outcome, only who captures the surplus.

5. Gaudium holds the right (cannot be held liable)

With no liability, Gaudium ignores the externality entirely and simply maximizes its own gross profit $\pi_G(g_G)$, which peaks at $g_G = 400$ (66,000). Silentium still independently chooses $g_S = 30$ (its optimal choice doesn't depend on g_G , only the profit level does).

Guests: Silentium = 30, Gaudium = 400 — more guests on Gaudium than the efficient 300.

Silentium's profit = 13,000 – 8,000 = 5,000. Gaudium's profit = 66,000. Total = 71,000.

Total profit (71,000) is lower than the efficient outcome (72,000) — a 1,000 deadweight loss from over-hosting on Gaudium, since it no longer bears the cost it imposes on Silentium. Note that if costless bargaining between the two hotels were possible, they could still reach the efficient 300-guest outcome even here: reducing Gaudium's guests from 400 to 300 costs Gaudium only 1,000 in forgone gross profit (66,000 – 65,000) but saves Silentium 2,000 in avoided losses (20×100), so Silentium could pay Gaudium something between 1,000 and 2,000 to voluntarily cut back — illustrating the core Coase-theorem insight that the initial assignment of rights affects only the distribution of surplus, not the efficient outcome, as long as bargaining is possible. The inefficient 400-guest result above reflects the case where no such bargain occurs.

Problem 8 — Igor's Fireworks

Supply: $Q(y) = 3$ (constant marginal cost of 3 per minute). Igor's demand: $x'(p) = 20 - p$, so

his inverse demand (marginal willingness to pay) is $p^I(x) = 20 - x$. Fireworks are a public good — non-rival, so everyone present enjoys the same duration simultaneously.

1. Igor alone: marginal WTP and Pareto-efficient duration

With a single consumer, efficiency requires marginal WTP = marginal cost: $20 - x = 3 \Rightarrow x = 17$ minutes. Igor's marginal willingness to pay for an additional minute, evaluated at the efficient duration, is $20 - 17 = 3$ (equal to marginal cost, as required).

2. Igor and Gertrude together: efficient duration and consumer surplus

Gertrude's demand $x^G(p) = 20 - 2p$ gives inverse demand $p^G(x) = 10 - 0.5x$. Because fireworks are a public good, efficiency requires the SUM of both individuals' marginal willingness to pay (not just one person's) to equal marginal cost — the Samuelson condition:

$$(20 - x) + (10 - 0.5x) = 3 \Rightarrow 30 - 1.5x = 3 \Rightarrow x = 18 \text{ minutes}$$

The efficient duration rises from 17 to 18 minutes once Gertrude is added — since the good is non-rival, an extra viewer adds pure value at no extra cost, so more of the good should be provided. Consumer surplus at the efficient quantity (area between the summed-WTP curve and marginal cost):

$$CS = \frac{1}{2} \cdot (30 - 3) \cdot 18 = \frac{1}{2} \cdot 27 \cdot 18 = 243$$

General rule: for a public good, the Pareto-efficient quantity is found where the vertical sum of all individual marginal-willingness-to-pay curves equals marginal cost ($\sum_i MWTP_i(x) = MC(x)$) — unlike a private good, where each individual's own price equals MC and quantities are summed horizontally across individual demands.

3. Igor covers all costs himself: consumer surplus

If Igor alone pays, he rationally provides fireworks only up to the point where HIS OWN marginal benefit equals cost — ignoring Gertrude's benefit, since he doesn't need her agreement or payment. This reverts to the single-consumer solution: $x = 17$ minutes (a smaller duration than the jointly efficient 18 minutes).

Igor's surplus (he pays the full cost): value from 0 to 17 minus cost = $195.5 - 3(17) = 195.5 - 51 = 144.5$. Gertrude pays nothing, so her entire value from watching is pure surplus: her value over 0 to 17 minutes is 97.75.

$$\text{Total CS} = 144.5 + 97.75 = 242.25 \text{ (versus 243 at the fully efficient } x = 18)$$

This is the classic public-goods free-rider problem: because Igor doesn't account for Gertrude's benefit when unilaterally deciding how much to fund, the fireworks display is under-provided relative to the efficient level, producing a small deadweight loss ($243 - 242.25 = 0.75$) even though Gertrude enjoys a costless windfall.

4. Cost-sharing: $p^G = \tau \cdot p$ for Gertrude, $p^I = (1-\tau) \cdot p$ for Igor

(a) $\tau = \frac{1}{2}$ — Gertrude's and Igor's demand, and interpretation

At $\tau = \frac{1}{2}$ each pays half of the overall price p : $p^I = 0.5p$ and $p^G = 0.5p$.

$$\text{Igor's demand: } x^I = 20 - 0.5p \quad \text{Gertrude's demand: } x^G = 20 - 2(0.5p) = 20 - p$$

Interpretation: at any common price p , Igor always wants more fireworks than Gertrude ($20 - 0.5p > 20 - p$ for any $p > 0$), because Igor's underlying valuation of fireworks is higher than Gertrude's (his marginal willingness to pay, $20 - x$, exceeds hers, $10 - 0.5x$, at every duration). Since the two must actually agree on the SAME shared duration x (it's a non-rival public good, not something they can consume in different amounts), an equal 50/50 split of the price does not produce matching demands — solving $x^I = x^G$ under $\tau = \frac{1}{2}$ only holds at the trivial price $p = 0$. In other words, a flat, equal cost-sharing rule fails to reflect the two consumers' differing valuations and cannot support a genuine (positive-price) shared equilibrium. A true Lindahl equilibrium instead requires personalized prices proportional to each person's actual marginal willingness to pay at the efficient quantity — here, at $x = 18$, Igor's MWTP is 2 and Gertrude's is 1 (summing to the marginal cost of 3), implying Igor should cover two-thirds of the cost ($\tau = 1/3$) rather than half.

Problem 9 — Institutions and Externalities

(Labeled "Problem 6" a second time in the original assignment — renumbered here to avoid confusion with the city-consultant problem above.)

1. The Williamson puzzle and its relation to the First Welfare Theorem

Oliver Williamson's puzzle asks: if competitive markets achieve efficient outcomes (per the First Welfare Theorem), why does so much economic activity happen inside firms — hierarchical organizations that replace market/price-based coordination with internal, non-market direction — rather than being coordinated entirely through arm's-length market contracting? The puzzle relates directly to the First Theorem because the theorem's efficiency result depends on strong, often-unrealistic assumptions: complete contracts, zero transaction costs, and full information. In reality, contracting, monitoring, and enforcement are costly, and information is imperfect — under these frictions, organizing certain transactions inside a firm (hierarchy) can be more efficient than coordinating them through the market. Firms exist precisely because real markets are not the frictionless, complete markets the theorem assumes.

2. Definition 6.1 — what does "institution" include?

Both markets and non-market institutions. Markets themselves are institutions — a market with well-defined property rights and enforceable contracts can internalize interdependencies through prices (e.g., a market for pollution permits). Non-market institutions include firms and internal hierarchies (Williamson's point — a firm internalizes certain interdependencies, like coordination between a supplier and manufacturer, through management rather than external contracting), as well as government regulation, the legal system (courts, property law), social norms and customs, and informal community governance arrangements (e.g., locally managed common-pool resources).

3. Examples of externalities in production and consumption

- Negative production externality: a factory's air or water pollution imposes health and cleanup costs on nearby residents and firms who are not compensated.
- Positive production externality: a firm's research and development generates knowledge spillovers that benefit other firms, which can learn from or build on the innovation without paying for it.
- Negative consumption externality: secondhand smoke, or noise from a loud neighbor, imposes a cost on people nearby who did not choose to be exposed.
- Positive consumption externality: getting vaccinated reduces disease transmission risk for others (herd immunity) — a benefit to third parties beyond the vaccinated individual.

5. Why some externalities can't be internalized by markets: climate change and nuclear power

Anthropogenic climate change: the affected parties include essentially all of humanity, spread across many sovereign nations with no overarching authority to define or enforce global property rights, and — critically — future generations who cannot participate in any present-day market transaction or bargain to protect their own interests. There is no way to buy, sell, or price emissions rights with people who do not yet exist, and no global government to assign and enforce the necessary worldwide property rights, so Coasean market solutions are not feasible at this scale.

Nuclear power: the central externality concern is catastrophic tail risk (a severe accident) and radioactive waste that remains hazardous for thousands of years — again extending far beyond the lifetime of any market participant or contract. Private insurance markets are generally unable to price such low-probability, catastrophically large, extremely long-horizon risks, and — as with climate change — future generations who may bear the consequences cannot negotiate compensation today. In both cases, markets fail to internalize the

externality because (a) affected parties include people who do not yet exist and cannot transact today, and (b) the sheer global scale and diffuseness of the affected population make Coasean bargaining prohibitively costly even among those alive now.